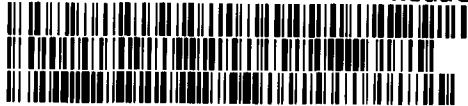


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TCGA-EM-A2CK-01A-PR

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## Surgical Pathology Consultation Report

ICD-0-3  
Carcinoma, papillary, follicular variant  
8340/3  
Site: thyroid, NOS C73.9 w/ 6/18/14

| Criteria                       | Yes        | No           |
|--------------------------------|------------|--------------|
| Diagnosis Discrepancy          |            | X            |
| Primary Tumor Site Discrepancy |            | X            |
| HIPAA Discrepancy              |            | X            |
| Prior Malignancy History       |            | X            |
| Dual/Synchronous Primary       |            | X            |
| Case is (Single):              | QUALIFIED  | DISQUALIFIED |
| Reviewed (Initials):           | w/ 6/18/14 |              |
| Date Reviewed:                 | 6/7/11     |              |

### Specimen(s) Received

1. Nck: RIGHT JUGULAR NODE-STITCH MARKS APEX
2. Neck:right posterior triangle - stitch marks apex
3. level 5A lymph nodes right neck
4. left lower parathyroid- QS
5. total thyroidectomy stitch mark upper right pole
6. right lower parathyroid - QS

### Diagnosis

1. Metastatic papillary thyroid carcinoma, classical variant, with extranodal extension: Lymph node (right jugular) biopsy
2. Metastatic papillary thyroid carcinoma, classical variant: Lymph nodes, 4/19 (right posterior triangle) dissection
3. No pathological diagnosis: Lymph nodes, 3 (right level 5A) dissection specimen
4. No pathological diagnosis: Parathyroid  
No pathological diagnosis: Thymus  
- (left lower neck) biopsy
5. Multifocal papillary carcinoma, dominant 2.3 cm, right, follicular variant; second 0.6 cm, right, classical variant: Thyroid  
Metastatic papillary thyroid carcinoma: Lymph node, 1 of 2, isthmic  
- Total thyroidectomy specimen
6. No pathological diagnosis: Parathyroid (right lower) biopsy

### Synoptic Data

|                     |                                                |
|---------------------|------------------------------------------------|
| Clinical History:   | No radiation exposure                          |
| Procedure:          | Total thyroidectomy with right neck dissection |
| Received:           | Fresh                                          |
| Specimen Integrity: | Intact                                         |
| Specimen Size:      | Right lobe: 4.5 cm<br>2.8 cm<br>2.2 cm         |

|                                 |                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------|
|                                 | Left lobe:                                                                                              |
|                                 | 4 cm                                                                                                    |
|                                 | 2.5 cm                                                                                                  |
|                                 | 1.5 cm                                                                                                  |
|                                 | Isthmus +/- pyramidal lobe:                                                                             |
|                                 | 4.8 cm                                                                                                  |
|                                 | 1.6 cm                                                                                                  |
|                                 | 0.8 cm                                                                                                  |
|                                 | Right neck dissection:                                                                                  |
|                                 | 20.2 cm                                                                                                 |
|                                 | 7 cm                                                                                                    |
|                                 | 3.5 cm                                                                                                  |
| Specimen Weight:                | 24.5 g                                                                                                  |
| Tumor Focality:                 | Multifocal, ipsilateral                                                                                 |
|                                 | Multifocal, midline (isthmus)                                                                           |
| ----- DOMINANT TUMOR -----      |                                                                                                         |
| Tumor Laterality:               | Right lobe                                                                                              |
| Tumor Size:                     | Greatest dimension: 2.3cm                                                                               |
|                                 | Additional dimension: 1.7 cm                                                                            |
|                                 | Additional dimension: 1.5 cm                                                                            |
| Histologic Type:                | Papillary carcinoma, follicular variant                                                                 |
|                                 | Follicular architecture                                                                                 |
|                                 | Classical cytomorphology                                                                                |
| Histologic Grade:               | Not applicable                                                                                          |
| Margins:                        | Margins uninvolved by carcinoma                                                                         |
| Tumor Capsule:                  | Partially encapsulated                                                                                  |
| Tumor Capsular Invasion:        | Not identified                                                                                          |
| Lymph-Vascular Invasion:        | Not identified                                                                                          |
| Extrathyroidal Extension:       | Not identified                                                                                          |
| ----- SECOND TUMOR -----        |                                                                                                         |
| Tumor Laterality:               | Right lobe                                                                                              |
| Tumor Size:                     | Greatest dimension: 0.6cm                                                                               |
|                                 | Additional dimensions: 0.4 cm                                                                           |
| Histologic Type:                | Papillary carcinoma, classical (usual)                                                                  |
|                                 | Classical (papillary) architecture                                                                      |
|                                 | Classical cytomorphology                                                                                |
| Histologic Grade:               | Not applicable                                                                                          |
| Margins:                        | Margins uninvolved by carcinoma                                                                         |
|                                 | Distance of invasive carcinoma to closest margin: 1mm                                                   |
| Tumor Capsule:                  | None                                                                                                    |
| Tumor Capsular Invasion:        | Present, extent widely invasive                                                                         |
| Lymph-Vascular Invasion:        | Not identified                                                                                          |
| Perineural Invasion:            | Not identified                                                                                          |
| Extrathyroidal Extension:       | Not identified                                                                                          |
| TNM Descriptors:                | m (multiple primary tumors)                                                                             |
| Primary Tumor (pT):             | pT2: Tumor more than 2 cm, but not more than 4 cm, limited to thyroid                                   |
| Regional Lymph Nodes (pN):      | pN1b: Metastases to unilateral, bilateral or contralateral cervical or superior mediastinal lymph nodes |
|                                 | Number of regional lymph nodes examined: 25                                                             |
|                                 | Number of regional lymph nodes involved: 6                                                              |
|                                 | Lymph Node, Extranodal Extension Present                                                                |
| Distant Metastasis (pM):        | Not applicable                                                                                          |
| Additional Pathologic Findings: | Parathyroid gland(s), within normal limits                                                              |
|                                 | Other: additional 2 papillary microcarcinomas in the isthmus, 0.1 and 0.3 cm                            |

Ancillary Studies:

Type: immunohistochemistry

Results: Dominant nodule - CK 19 positive, galectin negative, HBME 1 focally positive; second dominant nodule CK 19 positive, galectin positive, HBME 1 positive; largest metastatic lymph node - CK 19 positive, galectin positive, HBME 1 positive

\*Pathologic Staging is based on AJCC/UICC TNM, 7th Edition

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**Comment**

The dominant nodule stains for CK19 and focally for HBME-1 but is negative for galectin-3. In contrast, the second dominant nodule is strongly positive for CK19, galectin-3 and HBME-1, similar to the pattern seen in the metastatic lymph node. This immunoprofile and the morphology indicate that the metastasizing tumor is the second dominant nodule (widely invasive classical variant papillary carcinoma, 0.6 cm, right).

verified by:

Electronically

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**Gross Description**

1. The specimen labeled with the patient's name and as "right jugular node-stitch marks apex" consists of a piece of fibrofatty tissue and muscle oriented with a stitch marking the apex, measuring 5.7 x 2.7 x 1.5 cm received in 10% buffered formalin. One lymph node is identified in the area of the stitch; it measures 1.6 cm in diameter. It is solid to cystic on cut surface. The specimen is submitted as follows:

1A,B lymph node sectioned in toto.

2. The specimen labeled with the patient's name and as "right posterior triangle-stitch marks apex" consists of a piece of fibroadipose tissue that measures 10.5 x 2.3 x 1.5 cm oriented with a stitch marking the apex, received in 10% buffered formalin. Multiple lymph nodes are identified measuring from 0.3 cm to 3.5 cm in diameter. The largest is partly cystic on cut surface. The lymph nodes are submitted in toto from apex to opposite end as follows:

2A,B, multiple lymph nodes, each block

2C-F largest lymph node sectioned

2G multiple lymph nodes

2H two lymph nodes

2I one lymph node bisected

2J,K two lymph nodes each block

3. The specimen labeled with the patient's name and as "level 5A lymph nodes, right neck" consists of one piece of fibroadipose tissue that measures 3.5 x 2 x 1 cm received in 10% buffered formalin. Three lymph nodes measure from 0.2-cm to 2.2 cm in diameter. The specimen is submitted as follows:

3A one lymph node

3B,C one lymph node bisected each block.

4. The specimen labeled with the patient's name and as "left lower parathyroid qs" consists of a tiny piece of tissue that measures less than 0.1 cm in maximum dimension. It is frozen for intraoperative consultation.

4A frozen section block resubmitted.

5. The specimen labeled with the patient's name and as "total thyroidectomy, stitch marks right upper pole" consists of a thyroid gland that is oriented with a suture and weighs 24.5 g. The right lobe measures 4.5 cm SI x 2.8 cm ML x 2.2 cm AP, the left lobe measures 4.0 cm SI x, 2.5 cm ML x, 1.5 cm AP and the isthmus measures 4.8 cm SI x 1.6 cm ML x 0.8 cm AP. The external surfaces have fibrous adhesions. No parathyroid glands and/or lymph nodes are identified grossly. The external surface is painted with Silver nitrate. The right lobe contains two well delineated nodules, one upper and one lower, that measure 0.6 cm SI x 0.4 cm ML x 0.4 cm AP and 2.3 cm SI x 1.7 cm ML x 1.5 cm AP respectively. The lower pole of the left lobe contains a well delineated nodule that measures 0.3 cm SI x 0.4 cm ML x 0.4 cm AP. The remainder of the thyroid tissue is grossly unremarkable. Two sections of right lobe nodule, one section of the left lobe nodule and one piece of normal tissue are stored frozen. The remainder of the specimen is submitted as follows:

5A-L right lobe, superior to inferior

5M-Q isthmus

5R-Z left lobe, superior to inferior

6. The specimen labeled with the patient's name and as "right lower parathyroid qs" consists of a piece of tissue that measures 0.4 cm in maximum dimension. It is frozen for intraoperative consultation.

6A frozen section block resubmitted

#### **Quick Section Diagnosis**

1. Left lower parathyroid: cellular parathyroid tissue with piece of thymic tissue.

Slide received \_\_\_\_\_; reported \_\_\_\_\_ (seen by Dr. \_\_\_\_\_)

2. Right lower parathyroid: positive for parathyroid tissue.

Slide received \_\_\_\_\_; reported \_\_\_\_\_

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